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Whitepaper On

AIRA

Category: Artificial Intelligence

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Abstract

Financial exclusion remains a significant challenge in Bangladesh, particularly among individuals who have regular financial activity but lack conventional credit histories. This project proposes an AI-based alternative credit scoring system that evaluates financial behavior using consent-based, identity-verified data from mobile financial service transactions. The system analyzes patterns in income stability, savings, spending, transaction consistency, and seasonal behavior to generate an interpretable credit or trust score. It further incorporates anomaly detection, explainable assessment, and a lender-facing conversational AI assistant to support human lending decisions, while anchoring each score on a blockchain ledger so it remains tamper-evident and portable across multiple lenders.

1 Vision: Making Trust Visible

1.1 The Problem Statement

In Bangladesh, millions of economically active individuals remain credit invisible to the formal financial system. A tea-stall owner, street vendor, small shopkeeper, seasonal farmer, or home-based entrepreneur may earn regularly, save consistently, and pay bills on time, yet still be denied formal credit due to the absence of a salary certificate, credit card, or traditional banking history. This creates a fundamental paradox of financial inclusion:

> *"A person can be financially active without being financially visible."*

Approximately 70 million Bangladeshis are estimated to fall into the "credit invisible" category, while formal account ownership is only 43.3% of adults, leaving an estimated 98 million people financially excluded. At the same time, more than 239 million Mobile Financial Services (MFS) accounts generate billions of digital transactions annually. Yet much of this behavioral data remains fragmented, siloed, and underutilized for credit assessment. Existing alternative-credit initiatives demonstrate that behavioral lending is feasible, but some limitations remain:

- **Fragmented Trust:** Positive financial behavior with one provider is often not recognized across other lenders or platforms.
- **Limited Data Coverage:** Single-source models may overlook complementary signals available across multiple MFS providers.
- **Unclear Decisions:** Many systems provide an approve/reject outcome without clearly explaining the factors behind the assessment.
- **Exclusion of Irregular Earners:** Conventional models may inadequately assess seasonal workers, farmers, small vendors, and thin-history borrowers.

Core Problem: Bangladesh lacks a unified, comprehensive, explainable and inclusive credit-assessment system capable of securely transforming fragmented financial behavior into reliable credit profiles. Consequently, the system favors those who are already easy to score while leaving the truly credit-invisible population underserved.

1.2 The Solution: AIRA – An AI-Powered Trust Layer

We propose **AIRA**, an AI-powered alternative credit intelligence platform that bridges everyday financial behavior and formal credit assessment. With explicit user consent, AIRA combines legitimate behavioral signals such as valid MFS transactions. AI analyzes these signals to identify indicators of financial reliability, including payment discipline, income stability, saving behavior, transaction consistency and

legitimate seasonality. These signals contribute to an Alternative Trust Score, giving lenders an additional, evidence-based view of an applicant's creditworthiness.
AIRA goes beyond generating a score by providing an explainable trust layer between financial behavior and lending decisions. A lender-facing AI assistant can explain the factors behind an assessment, distinguish seasonal income from genuine instability, and provide contextual insights for evaluation. The AI supports the decision rather than replacing the human lender.

1.2.1 What Makes AIRA Different

- **Trust That Travels:** Financial credibility can extend beyond a single provider, reducing the need to rebuild trust when approaching another participating lender.
- **A Complete Financial Picture:** Multiple behavioral signals are combined to provide a broader view of the applicant than any single financial source can provide.
- **Decisions That Can Be Explained:** Lenders can examine the behavioral evidence behind an assessment through plain-language explanations rather than relying on a black-box score.
- **Credit That Understands Context:** Seasonal and thin-history earners are evaluated according to their financial context rather than automatically penalized for irregular income patterns.
- **Privacy by Design:** Explicit consent, responsible data use, privacy protection, and human oversight ensure that greater financial visibility does not compromise user control.

AIRA therefore shifts credit assessment from asking *"What formal financial history does this applicant have?"* to understanding *"What does this applicant's financial behavior demonstrate?"*
The result is not automated lending, but a more informed, contextual, and explainable basis for credit decisions, particularly for individuals whose financial responsibility is real but whose traditional credit footprint is limited.

2 Use Case

In Bangladesh, many small business owners, street vendors, farmers, and informal earners have regular financial activity but lack the formal credit history required by conventional lending systems. Common questions include:

- How can a lender assess an applicant with no conventional credit history?
- How can legitimate seasonal income be distinguished from financial instability?
- How can suspicious or manipulated transaction behavior be identified before lending?
- Which lenders and organizations can this applicant access today, and what would unlock more?

AIRA addresses this gap by transforming consent-based alternative financial data into an interpretable trust score and lender decision-support system. Key use cases are:

2.1 Alternative Credit Scoring

- Verifies applicant identity via an NID cross-check before scoring, enforcing exactly one verified identity (one NID) per applicant.
- Analyzes consent-based MFS transactions to derive income stability, payment consistency, spending, savings, and transaction-diversity patterns, generating a normalized trust score and risk tier for credit-invisible and thin-file applicants.

- Unlocks a tiered partner directory of lenders as the applicant's verified history strengthens — visible, motivating proof of progress without exposing the score itself.
- Gives NGOs, MFIs, and small lending organizations an additional credit signal beyond conventional financial records..

2.2 Fraud, Seasonal and Behavioral Analysis

- Checksum/QR-verifies MFS statements against the issuing provider, rejecting edited or fabricated uploads before scoring.
- Detects suspicious patterns — circular transfers, repeated counterparties, unusual concentrations, sudden pre-application balance spikes — and distinguishes legitimate seasonal income from real instability.
- Flags anomalous or insufficiently supported applications for human review rather than fully automated decisions.

2.3 Lender Chatbot and Explainable Decision Support

- Presents the applicant's trust score, risk tier, financial behavior, contributing factors, and detected anomalies to the authorized lender, who can ask natural-language questions such as:
  - Why is this applicant's score low?
  - Is the income instability seasonal?
  - What factors are affecting this applicant's risk?
- Provides evidence-grounded explanations and recommendations while the authorized human lender retains the final decision.

3 Existing Solutions

- **Traditional Credit Scoring:** Formal credit history and bank records, limiting access for credit-invisible applicants.
- **MFS and Digital Lending:** bKash, Nagad, and Rocket enable digital activity and lending within their own ecosystems.
- **Alternative Credit Scoring:** Dana Fintech, Agam, Tala, and Credolab use alternative data for scoring, but none show borrowers which lenders their score currently unlocks.

4 Risks and Challenges

AIRA handles sensitive financial data and supports lending decisions, making security, fairness, privacy, and regulatory compliance critical requirements.

4.1 Technical and Data Risks

- **Model Accuracy / Drift:** Changing behavior may reduce reliability. **Mitigation:** Validation and continuous monitoring.

- **Data Quality / MFS-API Dependency:** Missing or limited data, and reliance on authorized third-party access. **Mitigation:** Preprocessing, missing-data handling, and a modular multi-source data layer.

4.2 Fraud and Score-Gaming Risks

- **Artificial Transactions / Forged Statements:** Circular transfers or manipulated documents may inflate apparent strength. **Mitigation:** Transaction-network analysis plus checksum/QR verification against the issuing MFS provider, so an edited or fabricated bKash/Nagad statement is rejected before it ever reaches scoring.
- **Identity and Score Farming:** Duplicate or fraudulent scoring attempts. **Mitigation:** NID verification with only one verified NID permitted per applicant, alongside rate limits and anomaly detection.

4.3 Security, Privacy and Fairness Risks

- **Data Exposure / Digital Exclusion:** Sensitive histories and cash-dependent thin-file users. **Mitigation:** Encryption and access control, plus a separate thin-file pathway with human review.
- **Bias and Seasonal Misclassification:** Models may penalize legitimate patterns. **Mitigation:** Fairness evaluation, explainability, and human review.

4.4 Regulatory, Operational and Business Risks

- **Compliance and Automated Lending:** Consent and accountability concerns. **Mitigation:** Privacy-by-design, compliance review, and human final authority.
- **System Availability / Lender Adoption:** Outages and institutional hesitance toward AI scores. **Mitigation:** Monitoring and failover, plus explainability and pilot deployment.
- **LLM Hallucination:** Chatbot may generate unsupported explanations. **Mitigation:** Ground responses in SHAP factors and applicant records.

4.5 Summary Risk Matrix

Risks are rated **High** impact where a failure would be difficult to reverse or would directly undermine trust in the system - fraud, privacy, bias, and regulatory risk fall here. Risks rated **Medium** impact are ones where an existing mitigation (monitoring, fallback data sources, SHAP grounding, human final decision) bounds the damage before it becomes critical.

Table 1: Summary Risk Matrix for AIRA

Risk	**Likelihood**	**Impact**
Model Accuracy / Drift	Medium	Medium
Data Quality / API Dependency	High	Medium
Fraud and Score Gaming	High	High
Data Privacy	High	High
Model Bias / Fairness	Medium	High
Regulatory Compliance	Medium	High
Lender Adoption	Medium	Medium
LLM Hallucination	Medium	Medium

5 Architecture & Infrastructure

AIRA's architecture is designed to deliver secure, explainable, and tamper-resistant credit scoring by integrating several specialized modules. These modules work together to turn a borrower's mobile banking history into a trust score, explain that score to lenders in plain language, and keep the entire process verifiable and fraud-resistant, all of which is tailored to the unbanked population of Bangladesh.

5.1 Main Features

- **User & Identity Management:** Secure registration, authentication, and session handling for three roles — borrowers, lenders, and administrators. Borrowers submit their National ID (NID) during onboarding for identity verification, while lender organizations are vetted manually by administrators. This ensures secure access control so each party sees only what it is permitted to.
- **AI Credit Scoring & Explainability:** Borrowers upload their bKash/Nagad transaction statements, from which the system engineers behavioral features and computes a trust score using a machine-learning model. Every score is decomposed into human-readable factor categories, so lenders receive an explanation rather than an unexplained number.
- **Dual Conversational Assistants:** A lender-facing chatbot explains why a score is what it is and supports the loan decision, while a borrower-facing chatbot provides only limited, high-level improvement tips. This protects the scoring model from being reverse-engineered while still guiding the user.
- **Fraud Control & Score Portability:** Suspicious activity is flagged for administrator review and bad actors can be blocked, while each finalized score is anchored on a blockchain ledger so it can be trusted and reused across multiple lenders without a central authority owning the data.

5.2 Main Components

- **Frontend (React.js):** Provides the user interface for three separate experiences — the Borrower App, the Lender App, and the Admin Panel. The Borrower App handles NID and statement upload, score progress (badge and category summary), the borrower chatbot, a partner directory, consent controls, and notifications. The Lender App handles applicant lookup, the full score-and-tier view, the lender "Coach" chatbot, and usage/billing. The Admin Panel handles borrower KYC, lender verification, fraud review, and audit oversight. The frontend is optimized for low-bandwidth use to remain accessible in rural areas.
- **Backend (Express.js):** Implements the server-side logic and secure REST endpoints for authentication, consent management, and data processing. It handles statement verification (checksum/QR), transaction parsing and feature engineering, and orchestrates calls to the AI/ML layer for scoring. It exposes a Lender API (score-query endpoint) for participating NGOs and lending organizations, runs the reminder/streak scheduler, performs blockchain anchoring, and manages fraud/anomaly flagging.
- **Database (PostgreSQL):** Securely stores user profiles, uploaded statements, parsed transactions, score history, tiers, consent logs, and lender query records. It retains historical score records needed for borrower progress tracking and for lenders reviewing how an applicant's profile has changed over time.
- **AI/ML Layer:**
  - **Scoring Model (XGBoost / LightGBM):** A gradient-boosted decision tree model that computes the trust score from engineered features, kept deterministic and reproducible for lender trust and regulatory review.

- **SHAP Explainer:** Attributes the score to individual features and groups them into readable factor categories.
- **LLM Orchestration (LangChain + GPT-5.6 Luna):** Powers both chatbots, grounded on the SHAP explanation so the language model only explains results and never computes the score. The layer is model-agnostic, so any comparable model can be substituted.

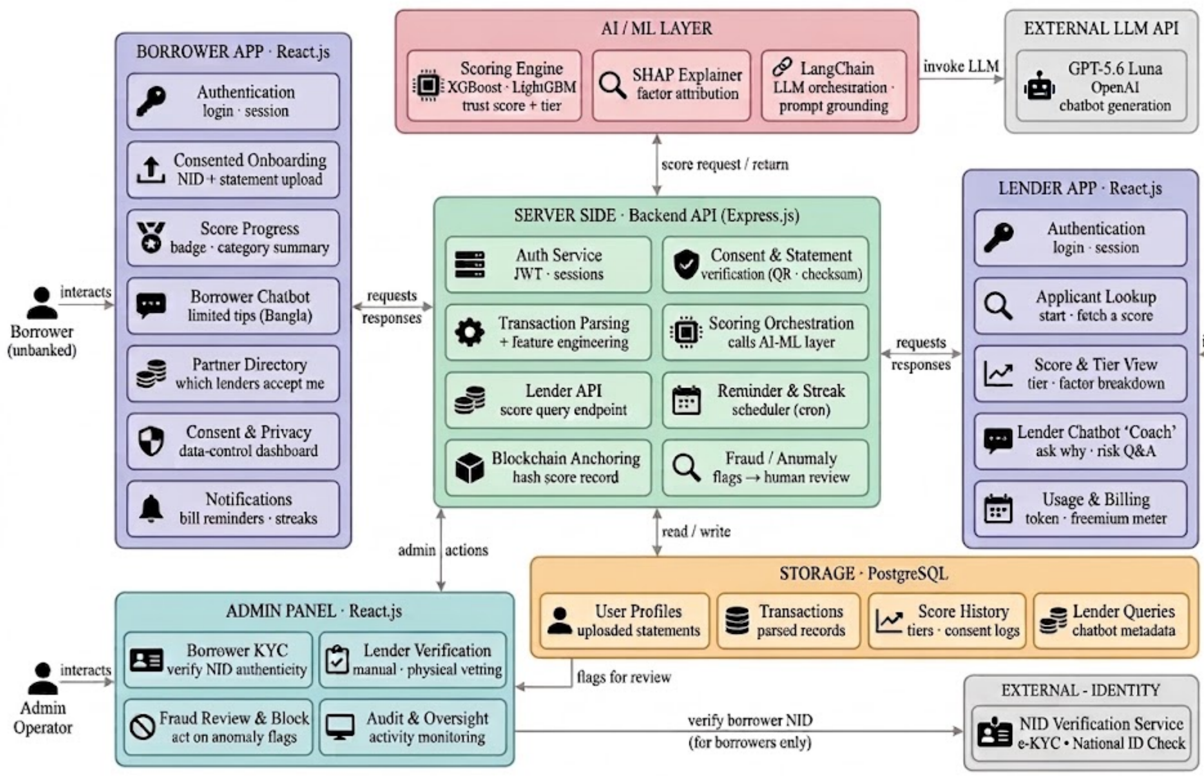

Figure 1: AIRA System Architecture

External Services:

- **NID Verification Service (e-KYC):** Validates borrower National IDs against the national database; a paid, regulated service that the prototype simulates.
- **GPT-5.6 Luna (OpenAI):** The external large-language-model API invoked for chatbot responses, chosen for its low cost, speed, and strong instruction-following, and configurable through LangChain.

5.3 Main Interactions

- **Borrower Journey:** Registers and verifies identity by submitting their NID, then gives consent and uploads a bKash/Nagad statement with a minimum of six months of history. The borrower views a simplified tier (Bronze/Silver/Gold/Platinum) and category summary, receives limited improvement tips from the borrower chatbot, and gets bill reminders and streaks.
- **Lender Journey:** A verified lender looks up an applicant and fetches the score, then views the full tier and factor breakdown. The lender uses the "Coach" chatbot to ask why a score is low or what loan amount is reasonable, and makes the final decision with real context rather than a bare number.
- **Backend & AI Operations:** Validates sessions, verifies statements, and engineers features from raw transactions, then calls the scoring model. SHAP produces the explanation, which is passed as

grounded context to GPT-5.6 Luna for both chatbots. Finalized scores are hashed and anchored on the blockchain ledger for cross-lender portability.

- **Admin Operations:** Verifies borrower NIDs through the e-KYC service and vets lender organizations manually. Reviews fraud/anomaly flags raised by the backend and blocks bad actors, providing human oversight over the automated pipeline.

6 Revenue & Distribution

6.1 Revenue Model

AIRA employs a diversified, token-based monetization strategy that keeps basic access affordable for small lenders while scaling naturally with usage.
Freemium Model:

- **Free Tier:** A basic trust-score check is enough for a simple approve/reject decision on a small loan and is available at no cost, encouraging broad adoption among NGOs, microfinance institutions (MFIs), and small lending organizations.

- **Paid Tier (Token-Based):** Lenders who need deeper capabilities like the lender-facing "Coach" chatbot analysis, seasonal-adjusted scoring, and bulk/high-volume checking pay using tokens or credits that scale with usage. An organization running many checks in a month buys tokens in bulk and becomes recurring revenue, while a small lender checking a few borrowers pays almost nothing.

  The token model is deliberately modeled on how modern AI platforms price access — pay-as-you-use — so cost always tracks value, and smaller organizations are never priced out. Because the prototype itself runs at near-zero cost on an open-source stack with a low-cost language model, this model comfortably sustains the platform as adoption grows.

6.2 Distribution Strategy

To reach both lenders and unbanked borrowers across Bangladesh, AIRA focuses on the community-level organizations that already serve this population:

- **NGO & Microfinance Partnerships:** Partner with NGOs, MFIs, and cooperatives already lending to rural and low-income communities. These organizations are both AIRA's paying customers and its channel to reach credit-invisible borrowers.

- **Local Lending Organizations:** Onboard small, community-based lenders and samitis that currently rely on informal judgment, giving them a structured, explainable score so they can lend more confidently.

- **Mobile Financial Service Collaboration:** Pursue data-access arrangements with bKash and Nagad under the national open-finance and digital-payment framework, enabling smoother onboarding through trusted platforms.

- **Pilot & Beta Access:** Offer free pilot access to a small group of consenting lenders and borrowers to validate accuracy, gather feedback, and build credibility before wider rollout.

- **Regulatory Engagement:** Align early with national open-finance and e-KYC frameworks to build the compliance credibility that institutional adoption requires.

7 Conclusion

AIRA addresses a real problem in Bangladesh, where millions of capable people are shut out of formal lending simply because they have no traditional financial record. By reading the mobile money activity people already generate through mobile banking like bKash and Nagad, AIRA turns everyday behavior into a fair, explainable trust score that lending organizations can act on with confidence.
Its strength lies in balancing intelligence with trust. A machine-learning model produces the score, while clear factor breakdowns keep it understandable rather than a black box, and identity verification, fraud review, and blockchain-anchored portability keep it honest. By giving lenders a transparent basis to decide and borrowers a way to build a financial reputation, AIRA offers a practical path toward financial inclusion, one trusted score at a time.